

William Muehlhausen

Engineering Software Developer |
CAD/PLM Automation | Data & Process
Engineering

Contact Information

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Technical Skills

Engineering & CAD Tools

- CATIA (Expert, KBE)
- ENOVIA (LCA / PLM integration)
- SolidWorks, Onshape
- GD&T (ASME Y14.5)

CAD/PLM Automation

- VB.NET (CATIA automation)
- C#
- VBA (Excel + CAD workflows)
- Python (engineering tools, scripting)
- BOM automation, validation tools

Data & Integration

- SQL Server, SSIS, ETL pipelines
- API / system integration (MES, SAP, Oracle, etc.)
- Data modeling & schema design
- Power BI

DevOps & Tools

- Azure DevOps (Repos, Pipelines)
- Git version control

Education

Data Science Nano-Degree

Udacity
Completed: June 2023

B.S. Mechanical Engineering

Washington State University December 2017

- Passed FE Exam and 3.75 Final Semester GPA

My career has progressed from a strong foundation in Tooling Design Engineering to a specialization in Software Development for Engineering Process Automation and Data Engineering. By combining my background in design engineering with software expertise, I focus on streamlining workflows and driving innovation through automation and data-driven solutions.

Career Path

Boeing - 7 years

Jun'22 - Aug'25 Tooling Knowledge Based Engineering - **Developer**

Jun'21 Accepted into Engineering **Rotation Program**

- ESRC - Manufacturing Engineer - **Python** 6 months
- IAS - NC Programming 6 months

Nov'19 Promoted **Lvl 2** Tooling Engineer

Aug'18 Hired **Lvl 1** Tooling Engineer

Electroimpact - 7 months

2016 and 2017 Summer Mechanical Engineering Internships

Employment

The Boeing Company

CAD/PLM Software Developer

June 2022 -August 2025

- Developed VB.NET applications to automate CAD and PLM workflows across CATIA and ENOVIA, reducing manual engineering effort and improving release accuracy
- Built a design validation tool that programmatically checks CATIA models/drawings and ENOVIA metadata against release requirements
- Created an automated BOM generation system extracting data from CATIA models into drawings and Excel for manufacturing use
- Integrated CAD/PLM systems with MES, SAP, and enterprise databases to enable end-to-end engineering data flow
- Partnered with engineering, IT, and manufacturing stakeholders to design solutions aligned with production workflows and system constraints
- Designed and implemented ETL pipelines (Python, SQL Server, SSIS) to support engineering applications and downstream data systems
- Modernized legacy engineering databases and applications (C#, SQL Server, Access), improving performance, scalability, and maintainability
- Collaborated with IT to gain access to restricted enterprise systems and enable new data integrations for engineering tools
- Led coordination with offshore developers to define backend architecture, relational data models, and application workflows
- Implemented version control and CI/CD practices using Azure DevOps (Git, Pipelines) to improve reliability and traceability of deployments
- Applied LLM-based techniques to extract structured engineering data from unstructured text fields

Electrical Systems Responsibility Center - Manufacturing Eng

Dec 21- June 22

- Collaborated as a team member in developing a ground-up production system for creating instructions for assembling wire bundles
- Developed software tools in VBA and Python to assist creating plans

Integrated Aero Structures - NC Programming

June 2021-Dec 2021

- Use CATIA Advanced Machining for extrusion machines that manufacture mass produced metal airplane structural components

Interior Responsibility Center - Tooling Engineer

August 2018 – May 2021

Design/Liaison

- Using CATIA V5, design variety of interior tool types throughout entire product lifecycle
 - Production ready assembly jigs, router fixtures, structural weldments

Core Competencies

CAD (CATIA, Solidworks, Fusion 360, Onshape)

Data Engineering

CATIA Automation

Data Warehousing

ETL, NLP and ML Pipelines

Machine Learning

AI

Data Visualization

Geospatial Data Analysis

Data Science

Leadership and Interests

- **Sierra Nevada Red Fox Research Technician**
 - Deployed non-invasive surveys including camera trapping and scat collection in the Oregon Cascades for fox occupancy modeling research
- **Competitive Cyclist** - Slalom Consulting - XC MTB, Road and Cross, Jerry Baker Memorial Velodrome
- **Outdoors** - Mountaineering, Ski Touring, and Backpacking
- **Racing Drone** - Purchased individual parts, built, soldered, and programmed first person view quadcopter

- and tools that work within automated systems on the production line with detailed part design on drawings using GD&T
- 3 years experience in applying Geometric Dimensioning and Tolerancing (GD&T) as per ASME y14.5 2009 and 2018
- Automated Panel Sanding Fixture
 - Designed pneumatic powered holding fixture for multiple 777X ceiling panel configurations as part of an automated sanding system
- No Stress Review Required Approver
 - Worked closely with stress engineering to get buyoff on tool designs
- Designed complex mold die surface and system using CATIA surfacing tools for emerging panel fabrication technology
- Designed interior tooling made from Thermwood LSAM 3D printer Designed in CATIA and sliced with Mastercam
- Built working relationships with different shops to effectively diagnose issues and work together as a team to fix issues
- Created a business process and programmed an analysis tool using excel VBA to be followed when a transportation cart is designed

Electroimpact

Mechanical Engineering Internships Summer 2016 and 2017

- Used Solidworks and CATIA for 777x aerospace tooling and worked on mechanical assembly and design revisions
- Developed procedures and designed tools with Solidworks for measuring and coring 4000+ holes under Boeing tooling specifications for 777x wings

Personal and Academic Projects

Personal Fitness AI Assistant

- Built an AI-powered fitness assistant that generates context-aware daily and weekly workout plans via automated email
- Integrated data from Strava, Google Calendar, weather APIs, and LLMs to enable context-aware training recommendations
- Designed and implemented REST API integrations and data pipelines to aggregate activity, schedule, and environmental data into a unified workflow
- Used LLMs to generate structured, personalized workout plans based on training history, availability, and real-world conditions, ensuring consistent and actionable output

CAD Projects (Onshape)

- Sliding, foldable vehicle-integrated camping bed system optimized for space efficiency, modularity, and rapid deployment
- Hardshell foldable rooftop tent incorporating hinge mechanisms and foam panel fiberglass construction for structural stiffness
- Elevated truck platform system designed for Trout Unlimited Beaver Management field operations

Whales, Shipping, and Sustainability Blog

- Utilized folium and geopandas in Jupyter Notebooks to visualize gray and blue whale sightings along the West Coast, showcasing their intersection with global shipping density. Used data analysis techniques to extract insights in Flask app

Disaster Response Pipeline

- Developed a natural language processing tool leveraging machine learning techniques, including scikit-learn, to effectively categorize real-world disaster data extracted from tweets and direct messages